

The Membrane Voltage

Neurons are excitable cells

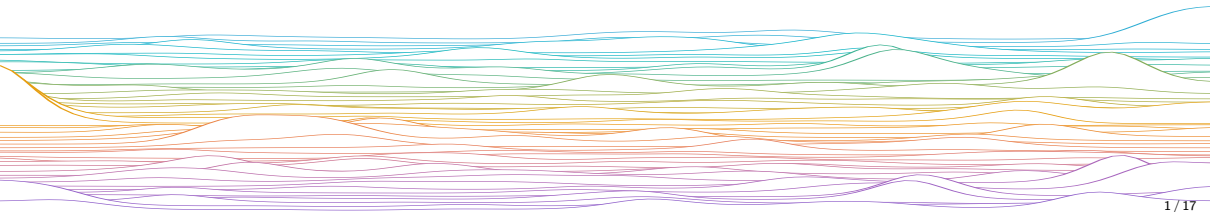
Computational Neuroscience

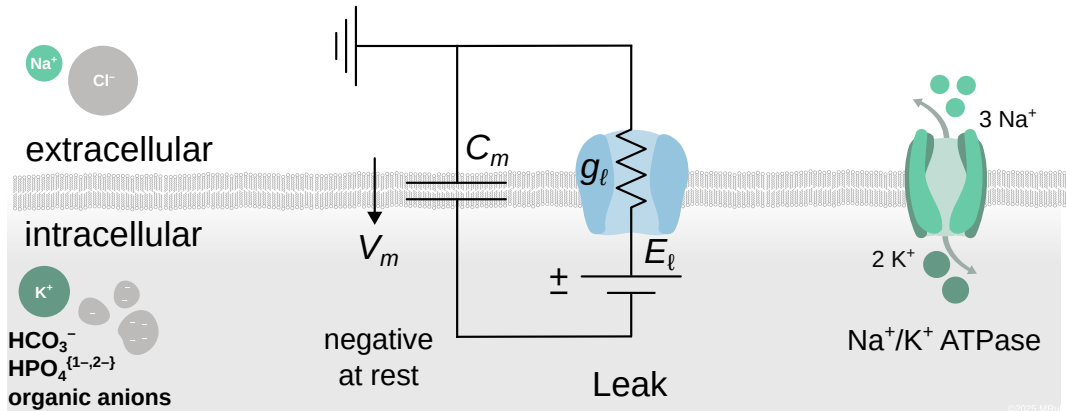
University of Bristol

M Rule

Learning outcomes:

- The effective RC circuit for passive membrane





Remember these?

$$I = \frac{d \text{ charge}}{d \text{ time}}$$

$$V = IR \quad \text{Ohm's law}$$

$$I = gV \quad \text{Ohm's law, conductance form}$$

$$\underline{I = C\dot{v}(t)} \quad \text{Definition of capacitance}$$

$$V = \text{pressure } \Delta$$

$$I = \text{water flow}$$

$$R = \text{resistance}$$

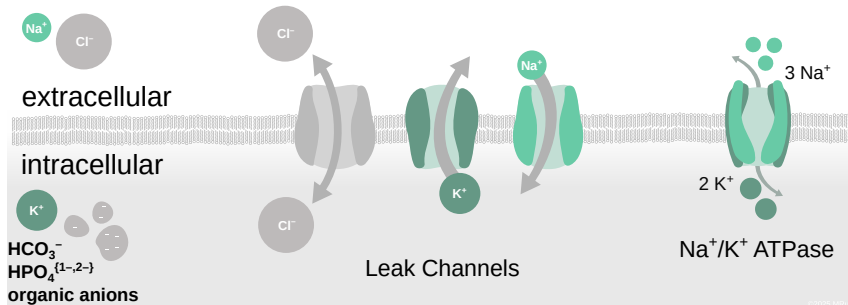
conductance

$$g = \frac{1}{R}$$

big g ~ big pipe

small ~ small





$$C_m \dot{v} = g_\ell (E_\ell - v)$$

$$C_m = 100 \times 10^{-12} \text{ Farads} \quad (100 \text{ picofarads})$$

$$E_\ell = -70 \times 10^{-3} \text{ Volts} \quad (-70 \text{ millivolts})$$

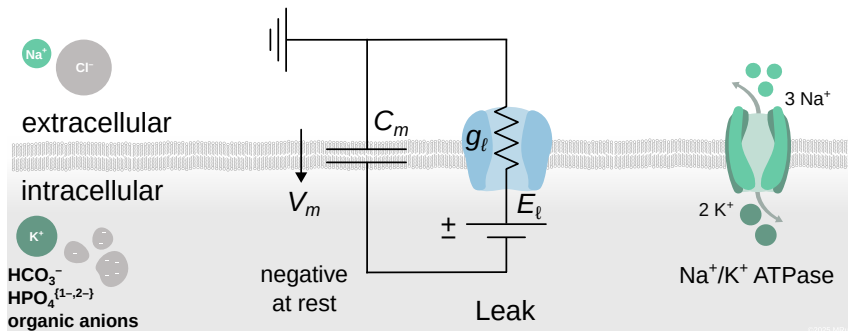
$$g_\ell = 10 \times 10^{-9} \text{ Siemens} \quad (10 \text{ nanosiemens})$$

cell membrane capacitance

resting potential

leak conductance

mho
1
ohm



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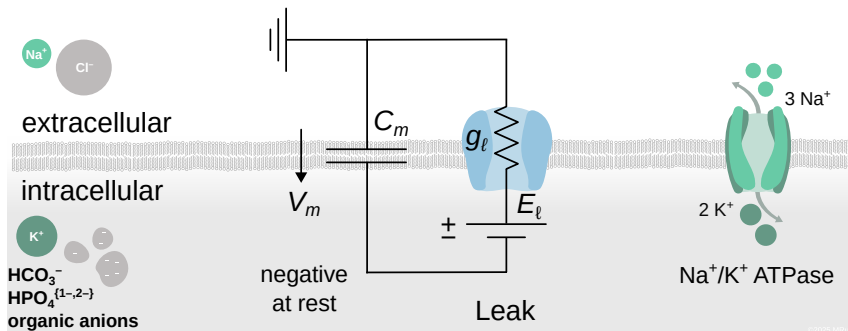
cell membrane capacitance

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resting potential

$$g_\ell = 10 \times 10^{-9} \text{ Siemens} \quad (10 \text{ nanosiemens})$$

leak conductance



$$C_m \dot{v} = g_\ell (E_\ell - v)$$

$$\tau_m \dot{v} = E_\ell - v$$

$$C_m = 100 \times 10^{-12} \text{ Farads} \quad (100 \text{ picofarads})$$

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$$g_\ell = 10 \times 10^{-9} \text{ Siemens} \quad (10 \text{ nanosiemens})$$

$$R_m = 1/g_\ell = 100 \times 10^6 \text{ Ohms} \quad (100 \text{ megaohms})$$

$$\tau_m = r_m C_m = 10 \times 10^{-3} \text{ seconds} \quad (10 \text{ milliseconds})$$

cell membrane capacitance

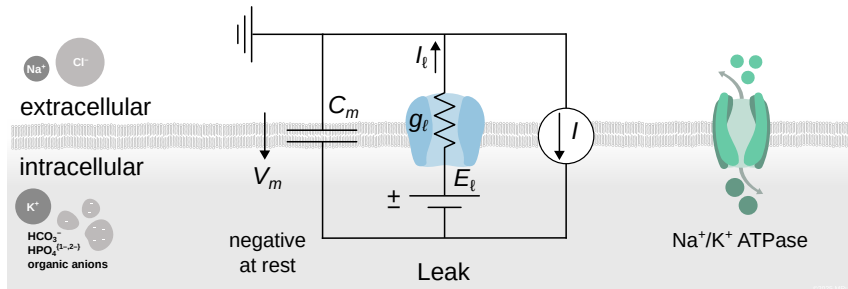
resting potential

leak conductance

membrane resistance

membrane time constant

Model: A neuron with applied current



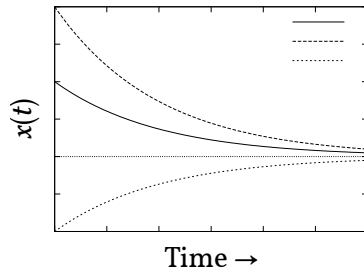
$$\tau_m \dot{v} = E_\ell - v + IR$$

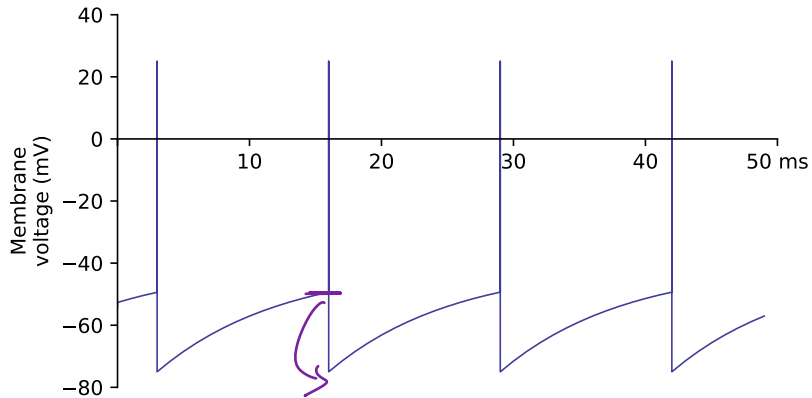
$$RC \dot{v} = E_\ell - v + IR$$

When is the previous model good? When is it bad?

Manually plot and solve some examples

Manually plot and solve some examples





Jargon

depolarize v_m increases

repolarize v_m decreases towards resting potential

hyperpolarize v_m is driven below resting potential

Sign conventions

Let $v_m = v_{\text{interior}} - v_{\text{exterior}}$ denote a neuron's membrane voltage

A positive current applied to a neuron makes v_m _____?

- ▶ **Engineer:** increase
- ▶ **Electrophysiologist:** decrease!

Compromise:

- ▶ **inward current** makes the voltage inside the cell more positive, relative to the outside
- ▶ **outward current** makes the voltage inside the cell more negative, relative to the outside

end

